

Getting Started with Imaclim Using Quarto

A Beginner's Guide to Documenting and Running Imaclim Models

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1 Introduction

1.1 What is Imaclim?

Imaclim is a macroeconomic model designed to analyze the interactions between the economy, energy systems, and climate policies. It is widely used for:

- Evaluating the economic impacts of climate policies (e.g., carbon taxes)
- Simulating energy transitions
- Producing long-term economic projections under different scenarios

Imaclim is typically implemented in **Scilab**, an open-source numerical computing environment.

1.2 What is Quarto?

Quarto is a modern, open-source scientific publishing system. Think of it as a powerful notebook that lets you:

- Write text (in Markdown) and code **in the same document**
- Produce outputs in multiple formats: **PDF**, **Word (.docx)**, **HTML**, and presentations
- Keep your analysis, results, and commentary all in one place

This tutorial will show you how to use Quarto to document your Imaclim runs, present results, and produce professional reports.

1.3 Prerequisites

Before you start, make sure you have:

1. **Scilab** installed — <https://www.scilab.org>
2. **Quarto** installed — <https://quarto.org>
3. **R** or **Python** installed (for post-processing results and producing charts)
4. The **IMACLIM-Country** repository cloned locally from GitHub: https://github.com/GaelleLeTreut/IMACLIM-Country/tree/SNBC3_run3_122025

Important: This tutorial is based on the branch **SNBC3_run3_122025**. Make sure you are on this branch before proceeding.

1.4 Key Documentation

Two essential references for understanding the model:

- **Gaëlle Le Treut's thesis** (most up-to-date summary of IMACLIM-S): <https://hal.science/tel-01707559>
 - **IMACLIM equations reference document**: https://www.centre-cired.fr/wp-content/uploads/2020/09/cired_wp_2020_85_le_treut.pdf
-

2 Setting Up Your Project

2.1 Repository Structure

Once cloned, the IMACLIM-Country repository is organised as follows:

IMACLIM-Country/

```
code/                                # Scilab scripts to run the model
  ImaclimS.sce                        # Main entry point - run this to launch a simulation
  Macro_Framework.sce                # Macro variables framework
  outputs_full_template.sce          # Saves output variables
  Test_AllConfig/
    main.sce                          # Creates dashboards with variants
    main1.sce                         # Runs all variant simulations
    def_of_MacroIncFRA.sce            # Declares variants to test
    Class_default_dashboard.sci

study_frames/                         # All model input variables
  study_frames_FRA2018/
    Dashboard_FRA2018.csv             # Main dashboard (entry point for variables)
    Country_Selection.csv             # Select which country to study
```

```

    AMSrun3mixnote/          # Scenario-specific input matrices
    IOT_Qtities_AMSrun3mixnote_1.csv # Inputs for 2030
    IOT_Qtities_AMSrun3mixnote_2.csv # Inputs for 2040
    IOT_Qtities_AMSrun3mixnote_3.csv # Inputs for 2050

Library/
    economic_equations.sci  # All model equations

outputs/                    # Results written here after each run
    AMSrun3mixnote/
        FullTemplate_AMSrun3mixnote.csv # All variables, all years
        Summary_AMSrun3mixnote.csv      # Key variables only

```

2.2 Creating Your First Quarto Document

Create a new file called `imaclim_report.qmd`. The `.qmd` extension stands for **Quarto Markdown**. Every Quarto document starts with a **YAML header** (the block between the `---` lines) that controls the document's settings:

```

---
title: "Imaclin Scenario Analysis"
author: "Your Name"
date: today
format:
  pdf:
    toc: true
  docx:
    toc: true
---

```

To **render** your document to PDF and Word, run this command in your terminal:

```
quarto render imaclim_report.qmd
```

3 Understanding the Imaclin Model Structure

3.1 Key Concepts

Before running scenarios, you should understand the three main components of Imaclin:

1. **Calibration** The model is calibrated to a **base year** (2018 for France). This means all model parameters are set so that the model exactly reproduces observed economic data for that year. You can skip re-calibration on subsequent runs by setting `skip_calibration;True` in the dashboard — this saves significant computation time.

2. Exogenous Variables These are variables that are set *outside* the model — typically projections about the future that you feed in. Examples include:

- Population growth and labour productivity
- Household saving rates (VAR_saving)
- Energy and world commodity prices
- Carbon tax settings

Note: If a variable is not present in the resolution system, it is treated as exogenous by default.

3. Scenarios A scenario defines a particular set of assumptions and policy choices. The main scenario used in this tutorial is **AMScrun3mixnote**. Variants are created by changing specific variables (e.g., household consumption in the aviation sector).

3.2 The Dashboard

The **Dashboard** (`study_frames/study_frames_FRA2018/Dashboard_FRA2018.csv`) is the central control file for any simulation. It contains all the variables that drive the model. Here is an annotated excerpt of the most important variables:

```
/// Simulation ///
Scenario;AMScrun3mixnote      // Name of the scenario
Nb_iter;3                     // Number of iterations (2018 to 2050)
skip_calibration;True         // Skip calibration to save time

/// Households ///
H_DISAGG;HH1                  // Number of household income groups (1-10 deciles)
VAR_saving;99                 // Variant on the household saving rate

/// Sector aggregation ///
AGG_type;AGG_23TME            // Use the 23-sector version of the model

/// Carbon tax ///
Carbone_ETS;False             // No ETS carbon pricing
ClosCarbRev;CstNetLend        // Carbon revenue recycling option

/// Productivity ///
Capital_Dynamics;%F           // No capital stock dynamics
Demographic_shift;True        // Productivity depends on demographics
Labour_product;True           // Productivity depends on active population

/// Energy & commodity prices ///
World_prices;True             // Import energy prices active (gas, oil...)

/// Variant simulations ///
study;SNBC3_RunChoices3       // RunChoice file used for variants
VAR_C_airtransp;              // Variant on aviation household consumption
```

3.3 The Reference Scenario

The reference scenario (**ref**) is always your baseline. All other scenarios are **compared against it** to measure the impact of a policy.

4 Running Your First Simulation

4.1 Step 1 — Select the Country

If this is your first time using the model, open the file `study_frames/Country_Selection.csv` and uncomment the country you want to study:

```
// For a study on France:
Country;France;FRA2018
```

4.2 Step 2 — Launch the Simulation

Open Scilab, navigate to the `code` folder, and run the main simulation file:

```
cd('IMACLIM-Country\code')
clear
exec ImaclimS.sce
```

This launches a simulation using the dashboard at: `study_frames/study_frames_FRA2018/Dashboard_FRA2018.csv`. Results are saved automatically to the `outputs/` folder.

4.3 Step 3 — Explore the Results

Once the simulation finishes, two output files are available for the scenario **AMRun3mixnote**:

File	Content
outputs/AMRun3mixnote/FullTemplate_AMRun3mixnote.csv	All simulation results (2018–2050)
outputs/AMRun3mixnote/Summary_AMRun3mixnote.csv	Summary results only
outputs/AMRun3mixnote/Time_2030/inputs	Input-output quantities for 2030
run_2030_AMRun3mixnote.csv	

5 Analysing an Economic Shock

One of Imacsim's core strengths is simulating the macroeconomic impact of a **shock** — a deliberate change to one or more input variables. This section walks through a concrete example: **analysing the macroeconomic impact of a -10% and -20% reduction in household consumption in the aviation sector.**

The aviation sector corresponds to row 19 of the household consumption vector: `Proj_Vol.C.val(19)`.

There are two ways to create a shock in Imacsim.

5.1 Method 1 — Shock via the Dashboard (recommended)

This is the preferred method as it keeps everything traceable and reproducible.

5.1.1 Step 1 — Declare the variants in the RunChoice file

Open `code/Test_AllConfig/SCNBC_RunChoice3.sce` and add the variant logic for your new variable:

```
// Modify aviation household consumption (row 19 = aviation sector)
if Scenario == "AMScrun3mixnote"
  if VAR_C_airtransp == "ref"          // Reference case
    Proj_Vol.C.val(19) = Proj_Vol.C.val(19);

    elseif VAR_C_airtransp == "-10%"    // -10% aviation consumption
      Proj_Vol.C.val(19) = Proj_Vol.C.val(19) * 0.9;

    elseif VAR_C_airtransp == "-20%"    // -20% aviation consumption
      Proj_Vol.C.val(19) = Proj_Vol.C.val(19) * 0.8;
    end
end
end
```

5.1.2 Step 2 — Register the variants to test

Open `code/Test_AllConfig/def_of_MacroIncFRA.sce` and add your variable:

```
test_fra_macro.VAR_C_airtransp = ['ref', '-10%', '-20%'];
```

5.1.3 Step 3 — Add the variable to the default dashboard class

Open `code/Test_AllConfig/Class_default_dashboard.sci` and declare the variable:

```
default_dashboard = struct( ..  
    'Carbone_ETS' , 'False', ..  
    ...  
    'VAR_C_airtransp', '');
```

5.1.4 Step 4 — Add the variable to the main Dashboard CSV

Open `study_frames/study_frames_FRA2018/Dashboard_FRA2018.csv` and add:

```
VAR_C_airtransp;-20%
```

5.1.5 Step 5 — Add the variable to the output template

Open `code/outputs_full_template.sce` so the variable is saved in results:

```
["VAR_C_airtransp", VAR_C_airtransp];
```

5.1.6 Step 6 — Generate the dashboards

```
cd('IMACLIM-Country\code\Test_AllConfig')  
exec main.sce
```

This creates an intermediate file `code/Test_AllConfig/dashboards.dat` containing one dashboard per variant. You can inspect them in Scilab:

```
dashboards      // lists all dashboards  
dashboards(1)   // shows the first dashboard - check your changes are there
```

5.1.7 Step 7 — Run all variant simulations

```
exec main1.sce
```

This runs each dashboard in turn and saves results for each variant in the `outputs/` folder.

5.2 Method 2 — Manual Shock on Input Files

This method is quicker for one-off experiments but less traceable.

1. Go to `study_frames/study_frames_FRA2018/AMSRun3mixnote/`
2. Copy and rename the folder — e.g. `AMSRun3mixlow`
3. Rename **all files** inside the new folder to match the new scenario name (every filename must contain `AMSRun3mixlow`)
4. Open `IOT_Qtities_AMSrun3mixlow_1.csv` (projection for 2030) and manually edit cell **Y20** to reduce aviation consumption. Repeat for `_2` (2040) and `_3` (2050) if needed.
5. Update the main dashboard to point to the new scenario:

```
Scenario;AMSRun3mixlow
```

6. Run the simulation:

```
clear
exec ImaclimS.sce
```

5.3 Analysing the Shock Results

Once the simulation is complete, compare the variant outputs against the reference:

File	Use
<code>outputs/AMSRun3mixnote/FullTemplate_AMSrun3mixnote.csv</code>	Alternative variables before/after shock
<code>outputs/AMSRun3mixnote/Summary_AMSrun3mixnote.csv</code>	Key variables summary
<code>outputs/AMSRun3mixnote/Time_2050/ecoT-run_2050_AMSrun3mixnote.csv</code>	Agent net lending (deficit) for 2050

Tip: `code/Macro_Framework.sce` is an important file for understanding how macro variables interact — refer to it when interpreting shock results.

6 Analysing Results in Quarto

6.1 Loading Results

Inside your `.qmd` file, you can load and process your Imaclim output directly. Here is an example using **R**:

```

```{r}
#| label: load-data
#| echo: true

Load results from the reference scenario
results_ref <- read.csv("../outputs/results_ref.csv")

Quick look at the data
head(results_ref)
```

```

Or using **Python**:

```

```{python}
#| label: load-data-python

import pandas as pd

results_ref = pd.read_csv("../outputs/results_ref.csv")
print(results_ref.head())
```

```

6.2 Plotting GDP Over Time

A typical first plot is **GDP evolution** across time steps:

```

```{r}
#| label: fig-gdp
#| fig-cap: "GDP evolution under the reference scenario"

library(ggplot2)

ggplot(results_ref, aes(x = year, y = GDP)) +
 geom_line(color = "#2E75B6", linewidth = 1.2) +
 geom_point(color = "#2E75B6", size = 2.5) +
 labs(
 title = "GDP - Reference Scenario",
 x = "Year",
 y = "GDP (billion EUR, constant prices)"
) +
 theme_minimal()
```

```

6.3 Comparing Scenarios

To compare the reference scenario with a carbon tax scenario, you can overlay the two:

```

```{r}
#| label: fig-scenario-comparison
#| fig-cap: "GDP comparison: Reference vs Carbon Tax"

results_carbon_tax <- read.csv("../outputs/results_carbon_tax.csv")

Add a scenario label to each dataset
results_ref$scenario <- "Reference"
results_carbon_tax$scenario <- "Carbon Tax"

Combine
all_results <- rbind(results_ref, results_carbon_tax)

ggplot(all_results, aes(x = year, y = GDP, color = scenario)) +
 geom_line(linewidth = 1.2) +
 geom_point(size = 2.5) +
 scale_color_manual(values = c("Reference" = "#2E75B6",
 "Carbon Tax" = "#E05C2A")) +
 labs(
 title = "GDP: Reference vs Carbon Tax Scenario",
 x = "Year",
 y = "GDP (billion EUR, constant prices)",
 color = "Scenario"
) +
 theme_minimal()
```

```

7 Writing Your Report

7.1 Mixing Text and Results

One of Quarto's most powerful features is **inline code** — you can embed computed values directly in your text. For example:

```

The GDP in the final year of the reference scenario is
`r round(tail(results_ref$GDP, 1), 1)` billion EUR.

```

This will automatically insert the correct number when the document is rendered — no copy-pasting required.

7.2 Cross-References

You can reference figures and tables automatically:

```
As shown in @fig-gdp, the economy grows steadily
under the reference scenario.
```

7.3 Adding a Summary Table

```
```{r}
#| label: tbl-summary
#| tbl-cap: "Key indicators - Reference Scenario"

library(knitr)

summary_table <- data.frame(
 Indicator = c("GDP (final year)", "CO2 emissions (final year)",
 "Household consumption", "Investment"),
 Value = c("3,250 bn EUR", "320 MtCO2", "1,850 bn EUR", "620 bn EUR"),
 Unit = c("constant 2019 EUR", "MtCO2eq",
 "constant 2019 EUR", "constant 2019 EUR")
)

kable(summary_table)
```
```

8 Rendering Your Document

8.1 Render to Both PDF and Word

To render to both formats at once, simply run:

```
quarto render imaclim_report.qmd
```

Quarto will produce:

- imaclim_report.pdf
- imaclim_report.docx

8.2 Render to a Specific Format Only

```
# PDF only
quarto render imaclim_report.qmd --to pdf

# Word only
quarto render imaclim_report.qmd --to docx
```

8.3 Previewing While You Write

During drafting, use `quarto preview` to see a live-updating HTML version in your browser:

```
quarto preview imaclim_report.qmd
```

9 Tips and Good Practices

9.1 Keep Code and Documentation Together

The main advantage of Quarto is that your code, outputs, and commentary live in one file. If you re-run the model with updated data, simply re-render the document — all tables and figures update automatically.

9.2 Use `echo: false` for Clean Reports

If you want to show results but hide the code in the final report:

```
execute:
  echo: false
```

Or for a single chunk:

```
```${r}
#| echo: false
This code runs but won't appear in the report
```
```

9.3 Version Control

Put your `.qmd` files under **Git** version control. This lets you track changes to your analysis over time and collaborate with colleagues.

```
git init
git add .
git commit -m "Initial Imaclim Quarto tutorial"
```

10 Next Steps

Now that you have a working Quarto setup for Imaclim, here are some things to explore next:

1. **Add more scenarios** — create a function that loops over all scenarios and produces a combined comparison figure automatically.
 2. **Parameterize your document** — use Quarto parameters to run the same report template for different countries or time horizons.
 3. **Publish online** — use `quarto publish` to share your report as a website via GitHub Pages or Quarto Pub.
 4. **Automate the pipeline** — combine the Scilab run and the Quarto render in a single shell script so everything runs end-to-end with one command.
-

11 Resources

- Quarto documentation: <https://quarto.org/docs>
- Scilab documentation: <https://www.scilab.org/documentation>
- IMACLIM-Country GitHub repository: <https://github.com/GaelleLeTreut/IMACLIM-Country>
- Gaëlle Le Treut's thesis (IMACLIM-S reference): <https://hal.science/tel-01707559>
- IMACLIM equations document: https://www.centre-cired.fr/wp-content/uploads/2020/09/cired_wp_2020_85_le_treut.pdf
- ggplot2 (R): <https://ggplot2.tidyverse.org>
- Matplotlib (Python): <https://matplotlib.org>